Universität St.Gallen

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Computer Networks

Network Security

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From insight to impact

Agenda

1. Setup and goals
2. Symmetric encryption
3. Asymmetric encryption
4. Summary

Lecture goals

- **Provide** a general overview of encryption mechanisms.
- **Compare** symmetric (AES) and asymmetric (RSA) approaches.
- **Explain** session keys (Diffie-Hellman) and TLS.
- **Discuss** certificates and digital signatures.

Two-column comparison

Symmetric encryption

- Single shared key
- Fast, suitable for bulk data
- Key distribution is the hard part

Asymmetric encryption

- Public / private key pair
- Solves key distribution
- Orders of magnitude slower

In practice, **TLS** combines both: asymmetric cryptography establishes a shared session key, which is then used with a symmetric cipher for the data stream.

The hard part is key distribution, not the cipher.

"

The cloud is just somebody else's computer.

Anonymous

"

The best way to predict the future is to invent it.

Alan Kay

Takeaways

- Symmetric cryptography is fast; asymmetric solves key distribution.
- **Certificates** bind public keys to identities; **TLS** uses them to bootstrap trust.
- **Diffie-Hellman** gives forward secrecy even if long-term keys leak later.

Questions?

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